

Planning Time: 1.737 ms
Execution Time: 1.052 ms

тип индекса	обоснование индекса	код индекса
частичный	Filter: ((wl.delivered IS NULL) OR (NOT wl.delivered)) Rows Removed by Filter: 54 в подзапросе SELECT 1 FROM "Order" o2 JOIN OrderItem oi2 ON oi2.order_id = o2.order_id LEFT JOIN WaybillLine wl ON wl.item_id = oi2.item_id WHERE o2.client_id = c.client_id AND (wl.delivered IS NULL OR wl.delivered <> 'Y')	CREATE INDEX idx_waybillline_undelivered ON WaybillLine (item_id) WHERE delivered == False;
составной	join по client_id и сортировка по MAX (o.created)	CREATE INDEX idx_order_client_created ON "Order" (client_id, created DESC);
покрывающий	cte => 3 seq scan	CREATE INDEX idx_orderitem_covering ON OrderItem (order_id) INCLUDE (count, dest_id);

Sort (cost=557.92..558.17 rows=103 width=2260) (actual time=0.651..0.659 rows=10.00 loops=1)
Sort Key: (count(o.order_id)) DESC
Sort Method: quicksort Memory: 27kB
Buffers: shared hit=12
-> GroupAggregate (cost=59.39..554.47 rows=103 width=2260) (actual time=0.593..0.650 rows=10.00 loops=1)
Group Key: c.client_name, d.dest_name
Buffers: shared hit=12
-> Sort (cost=59.39..59.64 rows=103 width=2168) (actual time=0.462..0.473 rows=103.00 loops=1)
Sort Key: c.client_name, d.dest_name, oi.dest_id
Sort Method: quicksort Memory: 46kB
Buffers: shared hit=9
-> Hash Left Join (cost=52.05..55.94 rows=103 width=2168) (actual time=0.266..0.364 rows=103.00 loops=1)
Hash Cond: (o.order_id = ow.order_id)
Buffers: shared hit=9
-> Hash Left Join (cost=36.53..40.12 rows=103 width=2160) (actual time=0.183..0.266 rows=103.00 loops=1)
Hash Cond: (o.order_id = ic.order_id)
Buffers: shared hit=7
-> Hash Left Join (cost=32.69..35.97 rows=103 width=2152) (actual time=0.142..0.208 rows=103.00 loops=1)
Hash Cond: (c.client_id = o_1.client_id)
Buffers: shared hit=6
-> Hash Right Join (cost=12.80..15.44 rows=103 width=1636) (actual time=0.035..0.081 rows=103.00 loops=1)
Hash Cond: (o.client_id = c.client_id)
Buffers: shared hit=3
-> Hash Right Join (cost=1.90..4.23 rows=103 width=20) (actual time=0.016..0.042 rows=103.00 loops=1)
Hash Cond: (oi.order_id = o.order_id)
Buffers: shared hit=2
-> Seq Scan on orderitem oi (cost=0.00..2.03 rows=103 width=8) (actual time=0.002..0.008 rows=103.00 loops=1)
Buffers: shared hit=1
-> Hash (cost=1.40..1.40 rows=40 width=16) (actual time=0.010..0.010 rows=40.00 loops=1)
Buckets: 1024 Batches: 1 Memory Usage: 11kB
Buffers: shared hit=1
-> Seq Scan on "Order" o (cost=0.00..1.40 rows=40 width=16) (actual time=0.003..0.006 rows=40.00 loops=1)
Buffers: shared hit=1
-> Hash (cost=10.40..10.40 rows=40 width=1620) (actual time=0.015..0.015 rows=10.00 loops=1)
Buckets: 1024 Batches: 1 Memory Usage: 10kB
Buffers: shared hit=1
-> Seq Scan on client c (cost=0.00..10.40 rows=40 width=1620) (actual time=0.009..0.010 rows=10.00 loops=1)
Buffers: shared hit=1
-> Hash (cost=19.76..19.76 rows=10 width=520) (actual time=0.101..0.104 rows=10.00 loops=1)
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=3
-> Hash Right Join (cost=8.70..19.76 rows=10 width=520) (actual time=0.094..0.101 rows=10.00 loops=1)
Hash Cond: (d.dest_id = (mode()) WITHIN GROUP (ORDER BY oi_1.dest_id))
Buffers: shared hit=3
-> Seq Scan on destination d (cost=0.00..10.70 rows=70 width=520) (actual time=0.003..0.003 rows=10.00 loops=1)
Buffers: shared hit=1
-> Hash (cost=8.57..8.57 rows=10 width=8) (actual time=0.085..0.088 rows=10.00 loops=1)
Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=2
-> GroupAggregate (cost=7.68..8.57 rows=10 width=8) (actual time=0.061..0.086 rows=10.00 loops=1)
Group Key: o_1.client_id
Buffers: shared hit=2
-> Sort (cost=7.68..7.93 rows=103 width=8) (actual time=0.052..0.058 rows=103.00 loops=1)
Sort Key: o_1.client_id
Sort Method: quicksort Memory: 27kB
Buffers: shared hit=2
-> Hash Join (cost=1.90..4.23 rows=103 width=8) (actual time=0.016..0.036 rows=103.00 loops=1)
Hash Cond: (oi_1.order_id = o_1.order_id)
Buffers: shared hit=2
-> Seq Scan on orderitem oi_1 (cost=0.00..2.03 rows=103 width=8) (actual time=0.002..0.006 rows=103.00 loops=1)
Buffers: shared hit=1
-> Hash (cost=1.40..1.40 rows=40 width=8) (actual time=0.009..0.011 rows=40.00 loops=1)
Buckets: 1024 Batches: 1 Memory Usage: 10kB
Buffers: shared hit=1

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-> Seq Scan on "Order" o_1 (cost=0.00..1.40 rows=40 width=8) (actual time=0.002..0.004 rows=40.00 loops=1)
    Buffers: shared hit=1
-> Hash (cost=3.34..3.34 rows=40 width=12) (actual time=0.037..0.038 rows=40.00 loops=1)
    Buckets: 1024 Batches: 1 Memory Usage: 10kB
    Buffers: shared hit=1
-> Subquery Scan on ic (cost=2.54..3.34 rows=40 width=12) (actual time=0.024..0.031 rows=40.00 loops=1)
    Buffers: shared hit=1
    -> HashAggregate (cost=2.54..2.94 rows=40 width=12) (actual time=0.024..0.028 rows=40.00 loops=1)
        Group Key: orderitem.order_id
        Batches: 1 Memory Usage: 32kB
        Buffers: shared hit=1
        -> Seq Scan on orderitem (cost=0.00..2.03 rows=103 width=8) (actual time=0.002..0.006 rows=103.00 loops=1)
            Buffers: shared hit=1
-> Hash (cost=15.02..15.02 rows=40 width=12) (actual time=0.077..0.077 rows=40.00 loops=1)
    Buckets: 1024 Batches: 1 Memory Usage: 10kB
    Buffers: shared hit=2
-> Subquery Scan on ow (cost=14.22..15.02 rows=40 width=12) (actual time=0.064..0.071 rows=40.00 loops=1)
    Buffers: shared hit=2
    -> HashAggregate (cost=14.22..14.62 rows=40 width=12) (actual time=0.064..0.068 rows=40.00 loops=1)
        Group Key: oi_2.order_id
        Batches: 1 Memory Usage: 32kB
        Buffers: shared hit=2
        -> Hash Join (cost=11.12..13.45 rows=103 width=12) (actual time=0.030..0.049 rows=103.00 loops=1)
            Hash Cond: ((oi_2.article)::text = (p.article)::text)
            Buffers: shared hit=2
            -> Seq Scan on orderitem oi_2 (cost=0.00..2.03 rows=103 width=13) (actual time=0.002..0.006 rows=103.00 loops=1)
                Buffers: shared hit=1
            -> Hash (cost=10.50..10.50 rows=50 width=42) (actual time=0.022..0.022 rows=30.00 loops=1)
                Buckets: 1024 Batches: 1 Memory Usage: 10kB
                Buffers: shared hit=1
            -> Seq Scan on product p (cost=0.00..10.50 rows=50 width=42) (actual time=0.003..0.005 rows=30.00 loops=1)
                Buffers: shared hit=1
SubPlan 2
-> Hash Join (cost=5.22..7.38 rows=36 width=4) (actual time=0.072..0.096 rows=55.00 loops=1)
    Hash Cond: (oi2.order_id = o2.order_id)
    Buffers: shared hit=3
-> Hash Right Join (cost=3.32..5.38 rows=36 width=4) (actual time=0.047..0.065 rows=55.00 loops=1)
    Hash Cond: (wl.item_id = oi2.item_id)
    Filter: ((wl.delivered IS NULL) OR (NOT wl.delivered))
    Rows Removed by Filter: 54
    Buffers: shared hit=2
-> Seq Scan on waybillline wl (cost=0.00..1.83 rows=83 width=5) (actual time=0.004..0.007 rows=83.00 loops=1)
    Buffers: shared hit=1
-> Hash (cost=2.03..2.03 rows=103 width=8) (actual time=0.021..0.021 rows=103.00 loops=1)
    Buckets: 1024 Batches: 1 Memory Usage: 13kB
    Buffers: shared hit=1
    -> Seq Scan on orderitem oi2 (cost=0.00..2.03 rows=103 width=8) (actual time=0.003..0.009 rows=103.00 loops=1)
        Buffers: shared hit=1
-> Hash (cost=1.40..1.40 rows=40 width=8) (actual time=0.017..0.018 rows=40.00 loops=1)
    Buckets: 1024 Batches: 1 Memory Usage: 10kB
    Buffers: shared hit=1
    -> Seq Scan on "Order" o2 (cost=0.00..1.40 rows=40 width=8) (actual time=0.008..0.011 rows=40.00 loops=1)
        Buffers: shared hit=1
Planning:
    Buffers: shared hit=294 read=4 dirtied=2
Planning Time: 1.909 ms
Execution Time: 0.758 ms

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показатель	до	после
total cost	72546.77	557.92
execution time	1.052 ms	0.758
buffers	shared hit=21	12

запрос 2

Nested Loop (cost=32.04..40.57 rows=1 width=644) (actual time=0.632..0.684 rows=16.00 loops=1)
Buffers: shared hit=325
-> WindowAgg (cost=31.90..31.95 rows=1 width=44) (actual time=0.200..0.228 rows=16.00 loops=1)
Window: w1 AS (ORDER BY (date_trunc('month')::text, o.created)))
Storage: Memory Maximum Storage: 17kB
Buffers: shared hit=293
-> GroupAggregate (cost=31.90..31.93 rows=1 width=36) (actual time=0.192..0.216 rows=16.00 loops=1)
Group Key: (date_trunc('month')::text, o.created))
Buffers: shared hit=293
-> Sort (cost=31.90..31.90 rows=1 width=24) (actual time=0.179..0.183 rows=103.00 loops=1)
Sort Key: (date_trunc('month')::text, o.created)), o.order_id
Sort Method: quicksort Memory: 29kB
Buffers: shared hit=293
-> Nested Loop (cost=0.30..31.89 rows=1 width=24) (actual time=0.021..0.153 rows=103.00 loops=1)
Buffers: shared hit=287
-> Nested Loop (cost=0.16..31.24 rows=1 width=25) (actual time=0.013..0.068 rows=103.00 loops=1)
Buffers: shared hit=81
-> Seq Scan on orderitem oi (cost=0.00..2.03 rows=103 width=13) (actual time=0.003..0.007 rows=103.00 loops=1)
Buffers: shared hit=1
-> Memoize (cost=0.16..0.67 rows=1 width=16) (actual time=0.000..0.000 rows=1.00 loops=103)
Cache Key: oi.order_id
Cache Mode: logical
Hits: 63 Misses: 40 Evictions: 0 Overflows: 0 Memory Usage: 5kB
Buffers: shared hit=80
-> Index Scan using "Order_pkey" on "Order" o (cost=0.15..0.66 rows=1 width=16) (actual time=0.001..0.001 rows=1.00 loops=40)
Index Cond: (order_id = oi.order_id)
Filter: (((created >= date_trunc('year')::text, (CURRENT_DATE - '1 year'::interval))) AND (created < date_trunc('month')::text, (CURRENT_DATE + '1 mon'::interval))))
Index Searches: 40
Buffers: shared hit=80
-> Index Scan using product_pkey on product p (cost=0.14..0.62 rows=1 width=42) (actual time=0.001..0.001 rows=1.00 loops=103)
Index Cond: ((article)::text = (oi.article)::text)
Index Searches: 103
Buffers: shared hit=206
-> Index Scan using client_pkey on client c (cost=0.14..8.16 rows=1 width=520) (actual time=0.002..0.002 rows=1.00 loops=16)
Index Cond: (client_id = (mode() WITHIN GROUP (ORDER BY o.client_id)))
Index Searches: 16
Buffers: shared hit=32
Planning:
Buffers: shared hit=337
Planning Time: 0.539 ms
Execution Time: 0.779 ms

тип индекса	обоснование индекса	код индекса
частичный	отсеять лишние даты, т.к. даты более годовой давности не используются	CREATE INDEX idx_order_recent ON "Order" (created, client_id, order_id) WHERE created >= '2024-01-01';
составной	в запросе в первую очередь интересует диапазон дат	CREATE INDEX idx_order_created_client ON "Order" (created, client_id, order_id);
покрывающий	index Cond: ((article)::text = (oi.article)::text) Index Searches: 103 Buffers: shared hit=206	CREATE INDEX idx_product_article_weight ON Product (article) INCLUDE (weight);

Sort (cost=28.60..28.70 rows=40 width=644) (actual time=0.333..0.336 rows=16.00 loops=1)
Sort Key: (date_trunc('month')::text, o.created))
Sort Method: quicksort Memory: 26kB
Buffers: shared hit=4
-> Hash Join (cost=21.69..27.53 rows=40 width=644) (actual time=0.265..0.326 rows=16.00 loops=1)
Hash Cond: ((mode() WITHIN GROUP (ORDER BY o.client_id)) = c.client_id)
Buffers: shared hit=4
-> WindowAgg (cost=10.79..13.72 rows=40 width=44) (actual time=0.232..0.269 rows=16.00 loops=1)
Window: w1 AS (ORDER BY (date_trunc('month')::text, o.created)))
Storage: Memory Maximum Storage: 17kB
Buffers: shared hit=3
-> GroupAggregate (cost=10.71..13.12 rows=40 width=36) (actual time=0.213..0.245 rows=16.00 loops=1)
Group Key: (date_trunc('month')::text, o.created))
Buffers: shared hit=3
-> Sort (cost=10.71..10.97 rows=103 width=24) (actual time=0.180..0.186 rows=103.00 loops=1)
Sort Key: (date_trunc('month')::text, o.created)), o.order_id
Sort Method: quicksort Memory: 29kB
Buffers: shared hit=3
-> Hash Join (cost=4.38..7.27 rows=103 width=24) (actual time=0.080..0.153 rows=103.00 loops=1)
Hash Cond: ((oi.article)::text = (p.article)::text)
Buffers: shared hit=3
-> Hash Join (cost=2.70..5.03 rows=103 width=25) (actual time=0.052..0.081 rows=103.00 loops=1)
Hash Cond: (oi.order_id = o.order_id)
Buffers: shared hit=2
-> Seq Scan on orderitem oi (cost=0.00..2.03 rows=103 width=13) (actual time=0.004..0.010 rows=103.00 loops=1)
Buffers: shared hit=1
-> Hash (cost=2.20..2.20 rows=40 width=16) (actual time=0.033..0.033 rows=40.00 loops=1)
Buckets: 1024 Batches: 1 Memory Usage: 10kB
Buffers: shared hit=1
-> Seq Scan on "Order" o (cost=0.00..2.20 rows=40 width=16) (actual time=0.006..0.023 rows=40.00 loops=1)
Filter: (((created >= date_trunc('year')::text, (CURRENT_DATE - '1 year'::interval))) AND (created < date_trunc('month')::text, (CURRENT_DATE + '1 mon'::interval))))
Buffers: shared hit=1
-> Hash (cost=1.30..1.30 rows=30 width=9) (actual time=0.020..0.020 rows=30.00 loops=1)
Buckets: 1024 Batches: 1 Memory Usage: 10kB
Buffers: shared hit=1
-> Seq Scan on product p (cost=0.00..1.30 rows=30 width=9) (actual time=0.002..0.006 rows=30.00 loops=1)
Buffers: shared hit=1
-> Hash (cost=10.40..10.40 rows=40 width=520) (actual time=0.015..0.016 rows=10.00 loops=1)

Buckets: 1024 Batches: 1 Memory Usage: 9kB
Buffers: shared hit=1
-> Seq Scan on client c (cost=0.00..10.40 rows=40 width=520) (actual time=0.008..0.010 rows=10.00 loops=1)
Buffers: shared hit=1
Planning:
Buffers: shared hit=155 read=5
Planning Time: 1.080 ms
Execution Time: 0.418 ms

показатель	до	после
total cost	32.04	28.60.
execution time	0.779 ms	0.418 ms
buffers	shared hit=325	4